

Emanuele Bernardi

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Applied mathematician (PhD, 2026) with more than four years of experience in mathematical modelling and simulation of biological systems. Focused on modelling biological variability from individual-level processes to population-scale dynamics, using MATLAB for numerical analysis and simulation. Seeking to apply mechanistic modelling to PK/PD, QSP and model-informed drug development.

Core Competencies

- **Modelling and Simulation of Biological Systems:** Compartmental, multi-scale and structured-population ODE/PDE and integro-differential models; numerical analysis and simulation of heterogeneous biological systems
- **Model Calibration & Data Analysis:** Calibration and parameter estimation against real-world Italian national health-system data; Monte Carlo simulation of individual-based stochastic models; analysis of imaging data
- **Modelling Tools:** MATLAB (expert), R/RStudio (basic), Python (basic), Git, $\LaTeX$; experience developing numerical simulation workflows and adapting to new modelling and simulation tools
- **Multidisciplinary Collaboration:** Industry-academia project with Camelot Biomedical Systems, international research visit (Paris), joint work across mathematics and biology

Professional Experience

- **Politecnico di Torino / University of Turin** **Torino, Italy**
PhD Researcher *Nov 2022-Feb 2026*
 - Developed and analysed mathematical models of heterogeneous biological populations: structured population dynamics, individual-based stochastic models, multi-scale and compartmental epidemiological models
 - Derived ODE/PDE and integro-differential systems from individual-level agent-based processes, linking variability between individuals to population-scale dynamics; carried out asymptotic analysis and numerical simulation in MATLAB
 - Published 2 peer-reviewed papers from the PhD (Physica D; Studies in Applied Mathematics)
 - Taught Calculus for Engineering (BSc, Politecnico di Torino, 2024)
 - Presented contributed talks and a poster at 5 international workshops and summer schools
- **University of Pavia / Camelot Biomedical Systems** **Pavia, Italy**
Research Scholarship *Nov 2021-Oct 2022*
 - Held scholarship under the grant “Machine Learning for Industry 4.0 and Digital Healthcare”, working directly with an industry partner
 - Developed and calibrated compartmental epidemiological models against real-world Italian national health system data during the COVID-19 pandemic
 - Developed models for logistics and facility management, processing and analysing large-scale datasets

- **Université Paris Cité, IMJ-PRG** **Paris, France**
Visiting PhD Student *Feb 2025-Mar 2025*
 - Collaborated on extensions of phenotype-structured population models to co-evolutionary dynamics between populations and their environment

Education

- **Politecnico di Torino / University of Turin** **Torino, Italy**
PhD in Pure and Applied Mathematics (cum laude) *2022-2026*
 Thesis: "Phenotype-structured integro-differential models: derivation and qualitative analysis." Supervisors: Prof. Tommaso Lorenzi, Prof. Andrea Tosin.
- **IUSS Pavia** **Pavia, Italy**
Master in Science and Technology (100/100 cum laude) *2019-2022*
- **University of Pavia** **Pavia, Italy**
Master's Degree in Mathematics (110/110 cum laude) *2019-2021*
- **University of Pavia** **Pavia, Italy**
Bachelor's Degree in Mathematics (110/110 cum laude) *2016-2019*

Languages

- Italian (native), English (C1, IELTS 7.5), French (basic).

Publications

- E. Bernardi, T. Lorenzi, A. Tosin (2026). Derivation and quasi-invariant asymptotics of phenotype-structured integro-differential models. *Physica D: Nonlinear Phenomena*, 497, 135402. [DOI link](#)
- E. Bernardi, T. Lorenzi, M. Sensi, A. Tosin (2025). Heterogeneously Structured Compartmental Models of Epidemiological Systems: From Individual-Level Processes to Population-Scale Dynamics. *Studies in Applied Mathematics*, 155(2):e70091. [DOI link](#)
- E. Bernardi, L. Pareschi, G. Toscani, M. Zanella (2022). Effects of vaccination efficacy on wealth distribution in kinetic epidemic models. *Entropy*, 24(2):216. [DOI link](#)